



Chapter 37- WAN Architectures & Wireless Fundamentals



1. Introduction to WANs (Wide Area Networks)



What is a WAN?

- A **WAN (Wide Area Network)** is a network that covers a **large geographical area** 🌐
- It connects **multiple LANs (Local Area Networks)** across cities, countries, or even continents



Why WAN is used?

- To connect **offices, data centers, and branches** of a company
- Helps in **data sharing and communication** between distant locations



Internet vs WAN

- The **Internet** itself is a WAN
- But in real-world usage, WAN usually means:
 - A company's **private network connections**



WAN over Internet

- Companies use **VPN (Virtual Private Network)** to create **secure connections over public internet**
- Makes communication **private and encrypted**



Important Note

- Different WAN technologies exist
- Some are **old (legacy)** but still used in some countries

IN India Example

- Early internet infrastructure in India was supported by private efforts (like industrial pioneers funding connectivity)

2. WAN Over Dedicated Connection (Leased Line)

What is a Leased Line?

- A **leased line** is a **dedicated connection between two sites**
- It is like renting a **private road only for your company** 🚗

Features

- Uses **PPP or HDLC protocols**
- Provides **direct and secure communication**
- Installed via **ISP infrastructure (not direct cable end-to-end)**

Why Companies Don't Buy It?

- Very **expensive** 💰
- High **maintenance cost**
- Long **installation time**

Topology: Hub-and-Spoke

- **Hub** = main location
- **Spokes** = branch offices connected to hub

Example:

- Pune = Hub
- Delhi = Spoke
- Both connected via leased line through ISP network

Problem

- Slow speeds and high cost → now replaced by **Ethernet WAN & VPNs**
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3. WAN Over Shared Infrastructure (Internet VPN)

What is VPN?

- VPN creates a **secure tunnel between sender and receiver**
- Data travels in **encrypted format** 

Tunnel Concept (Easy Example)

- Like a **road tunnel**:
 - You can't see inside from outside 
- Same with VPN:
 - Data is hidden and protected

Before VPN

- Data travels as **plain text**
- Anyone can read it 

After VPN

- Data is **encrypted**
- No third party can read it 

Types of VPN

1. Site-to-Site VPN

Definition

- Connects **two sites (office to office)** over internet

How it Works

1. Data is **encrypted** 
2. Encapsulated with **new headers**
3. Sent through VPN tunnel

4. Receiver **decrypts** data

Key Points

- Tunnel exists between **two routers**
- End devices don't create VPN themselves

Example

- Office A router ↔ Office B router

Limitations

- Doesn't support **multicast/broadcast**
- So routing protocols like **OSPF** don't work directly

Solution

- Use **GRE over IPSec**

Real Tool Example

- Palo Alto firewall supports Site-to-Site VPN
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2. Remote-Access VPN

Definition

- Allows **individual users (laptop/mobile)** to connect to company network

Technology Used

- Uses **TLS (Transport Layer Security)**
(same as HTTPS )

How it Works

- Install VPN software (e.g., Cisco AnyConnect)
- Device connects to company router/firewall
- Creates secure tunnel

Real-Life Example

- Work From Home 
- Laptop connects securely to office network

Popular Tools

- Cisco AnyConnect
- Palo Alto GlobalProtect

Site-to-Site vs Remote-Access VPN

Feature	Site-to-Site 	Remote Access 
Use Case	Office to Office	User to Office
Technology	IPSec	TLS
Users	Multiple devices	Single device
Connection	Permanent	On-demand

Split Tunnel VPN

What is it?

- Only **important traffic goes through VPN**
- Other traffic goes through normal internet

Example

- Office server access → VPN
- YouTube, Games → Normal Internet

Benefits

- Saves **bandwidth** 
- Reduces **cost** 

- Improves performance ⚡
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4. MPLS VPN (Multi Protocol Label Switching)

What is MPLS?

- A WAN technology using **labels instead of IP addresses**

Devices

- **CE Router** → Customer Edge
- **PE Router** → Provider Edge
- **P Router** → Provider Core

How it Works

- PE router adds **labels** to data
- Data is forwarded using labels (not IP)

Types

◆ Layer 3 MPLS

- CE ↔ PE use routing protocols (e.g., OSPF)
- Routes are shared

◆ Layer 2 MPLS

- CE routers act like directly connected
- No routing with PE
- Same subnet

Benefit

- Faster and efficient routing
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5. Internet Connectivity

Types

- Leased Line
 - MPLS
 - DSL
 - Cable (CATV)
 - Fiber Optic (most popular now ⚡)
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6. Wireless Fundamentals

Standards

- Defined by **IEEE 802.11**
- Certified by Wi-Fi Alliance

Important

- Wi-Fi is a **brand name**, not a standard
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7. Wireless Network Challenges

Issues

1. All devices receive all signals 
2. Privacy concerns 
3. Collision problems

Solution

- Wired: CSMA/CD
 - Wireless: **CSMA/CA** (avoid collisions)
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8. Radio Frequency (RF)



How Wireless Works

- Electric current → antenna → electromagnetic waves



Key Terms



Frequency

- Cycles per second (Hz)

Unit Meaning

Hz 1 cycle/sec

kHz 1,000 cycles

MHz 1 million

GHz 1 billion



Period

- Time for one cycle
- Example: 4 Hz = 0.25 sec



9. Wi-Fi Frequency Bands



2.4 GHz

- Longer range 
- Lower speed 
- More interference 



5 GHz

- Short range
- High speed 
- Less interference 

Example

- Slow cycling → long distance (2.4 GHz)
 - Fast cycling → short distance (5 GHz)
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10. Channels

Concept

- Bands are divided into channels

◆ 2.4 GHz

- Use **1, 6, 11** (non-overlapping)

◆ 5 GHz

- Many non-overlapping channels

Design

- Use **honeycomb pattern** for AP placement
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11. Wi-Fi Standards

Standard	Frequency	Speed	Name
802.11	2.4 GHz	2 Mbps	-
802.11b	2.4 GHz	11 Mbps	-
802.11a	5 GHz	54 Mbps	-
802.11g	2.4 GHz	54 Mbps	-
802.11n	2.4/5 GHz	600 Mbps	Wi-Fi 4
802.11ac	5 GHz	6.9 Gbps	Wi-Fi 5
802.11ax	2.4/5/6 GHz	Very Fast	Wi-Fi 6



12. Signal Behavior (Very Important)



1. Absorption

- Signal converts into heat
- Weakens signal



Example: Wall absorbs Wi-Fi



2. Reflection

- Signal bounces off metal



Example: Elevator → weak signal



3. Refraction

- Signal bends in different medium



Example: Glass, water



4. Diffraction

- Signal bends around obstacles



Example: Signal reaches behind wall



5. Scattering

- Signal spreads in all directions



Example:

- Rain 🌧️
- Dust
- Trees 🌳



Final Summary

- WAN connects **long-distance networks**
- Leased lines = **expensive but dedicated**
- VPN = **secure communication over internet**
- MPLS = **label-based fast routing**
- Wireless uses **radio waves**
- 2.4 GHz = range, 5 GHz = speed
- Signal behavior affects **Wi-Fi performance**

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SUMMARY



WAN Basics

- WAN connects **multiple LANs over large distance** 🌐
 - Used by companies to connect branches, data centers, offices
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WAN Types



1 Leased Line

- Dedicated private connection 🔒
- Uses PPP / HDLC
- Expensive 💰



2 VPN (Internet-based WAN)

- Secure communication over internet 🔒
 - Uses encryption
 - Types:
 - Site-to-Site VPN 🌐
 - Remote Access VPN 💻
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MPLS WAN

- Uses **labels instead of IP** 🚀
 - Faster forwarding
 - Used by ISPs
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Wireless Fundamentals

- Based on **IEEE 802.11**
- Uses radio waves 📶

Frequencies

- **2.4 GHz** → Long range, low speed
- **5 GHz** → Short range, high speed

Signal Issues

- Absorption 
- Reflection 
- Refraction 
- Diffraction 
- Scattering 

Key Concept

 WAN = Connectivity

 Wireless = Mobility

CONCLUSION

 Modern networks use:

- VPN instead of costly leased lines
- MPLS for enterprise-level performance
- Wireless for flexibility and mobility

 Final understanding:

WAN connects the world 

Wireless connects users 

Q & A

 **Q1: What is WAN?**

 **Answer:**

A WAN (Wide Area Network) is a network that connects multiple LANs across large geographical areas like cities or countries.

 **Example:**

Company branches in Mumbai, Delhi connected via WAN.

 **Q2: Difference between LAN and WAN?**

Feature	LAN	WAN
Area	Small	Large
Speed	High	Lower
Cost	Low	High

 **Q3: What is a leased line?**

 **Answer:**

A leased line is a dedicated private connection between two locations.

 **Key points:**

- Always ON 
 - Secure 
 - Expensive 
-

? Q4: Why companies don't prefer leased lines today?

✓ Answer:

- High cost 💰
- Slow deployment
- Maintenance heavy

👉 Replaced by VPN & MPLS

? Q5: What is VPN?

✓ Answer:

VPN (Virtual Private Network) creates a secure encrypted tunnel over the internet.

👉 Data becomes unreadable to 3rd party 🔒

? Q6: What is tunneling?

✓ Answer:

Tunneling means encapsulating data inside another packet for secure transmission.

👉 Like a tunnel 🚂 → no one can see inside

? Q7: Types of VPN?

✓ Answer:

1. Site-to-Site VPN 🌐
 2. Remote Access VPN 💻
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? Q8: Difference between Site-to-Site and Remote Access VPN?

Feature	Site-to-Site	Remote Access
Use	Office ↔ Office	User ↔ Office
Device	Router	Laptop
Protocol	IPSec	TLS

? Q9: Why GRE over IPSec is used?

✓ Answer:

Because IPSec does not support multicast/broadcast.

👉 GRE adds support for routing protocols like OSPF.

? Q10: What is Split Tunnel VPN?

✓ Answer:

Only important traffic goes via VPN, rest uses internet.

👉 Saves bandwidth + improves performance 🚀

? Q11: What is MPLS?

✓ Answer:

MPLS is a WAN technology that forwards packets using labels instead of IP.

👉 Faster than traditional routing 🚀

? Q12: MPLS components?

✓ Answer:

- CE (Customer Edge)
- PE (Provider Edge)
- P (Provider Core)

? Q13: Difference between MPLS L2 and L3?

Type	Description
L2 MPLS	Same subnet
L3 MPLS	Routing used

? Q14: What is Wi-Fi?

✓ Answer:

Wi-Fi is a wireless networking technology based on IEEE 802.11 standards.

? Q15: Difference between 2.4 GHz and 5 GHz?

Feature	2.4 GHz	5 GHz
Range	Long 📶	Short
Speed	Low 🐢	High ⚡
Interference	High ❌	Low ✓

? Q16: What are Wi-Fi channels?

✓ Answer:

Channels divide frequency bands to avoid interference.

👉 2.4 GHz → 1, 6, 11

? Q17: What is CSMA/CA?

✓ Answer:

Collision Avoidance method used in wireless networks.

👉 Devices check before sending data

? Q18: Difference between CSMA/CD and CSMA/CA?

Feature	CSMA/CD	CSMA/CA
Use	Wired	Wireless
Method	Detect collision	Avoid collision

? Q19: What is absorption?

☒ Signal absorbed → becomes weak

? Q20: What is reflection?

☒ Signal bounces off metal

? Q21: What is diffraction?

☒ Signal bends around objects

? Q22: What is scattering?

☒ Signal spreads in multiple directions

 ◆ SCENARIO-BASED QUESTIONS

? Q23: Company wants secure communication over internet. What will you use?

☒ **Answer:**
VPN (IPSec or SSL VPN)

? Q24: User working from home needs access to office network?

✓ Answer:

Remote Access VPN

? Q25: Multiple branches need constant connection?

✓ Answer:

Site-to-Site VPN or MPLS

? Q26: Wi-Fi signal weak in office?

✓ Answer:

Check:

- Interference
 - Walls (absorption)
 - Channel overlap
-

? Q27: How to improve Wi-Fi performance?

✓ Answer:

- Use 5 GHz
 - Proper channel planning
 - Add more APs
 - Avoid interference
-

? Q28: Why only channels 1, 6, 11 used?

✓ Answer:

Because they do not overlap → reduces interference

? Q29: What happens if too many users on same AP?

✓ Answer:

- Congestion
 - Packet loss
 - Low speed
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🎯 🔥 One Liner

“Explain WAN & Wireless in simple way”

✓ Best Answer:

WAN connects networks over long distance using technologies like VPN and MPLS, while wireless allows devices to connect without cables using radio frequencies like 2.4 GHz and 5 GHz. Both are essential for modern networking—WAN for connectivity and wireless for mobility.

🎯 ✓ FINAL TAKEAWAY

- ✓ WAN = Long distance communication 🌐
 - ✓ VPN = Secure communication 🔒
 - ✓ MPLS = Fast routing 🚀
 - ✓ Wireless = Flexible connectivity 📶
 - ✓ QoS + Wireless + WAN = Modern Network Backbone **100**
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